TSM 099 Quadratic and Rational Equations Test

Practice 1 | Version 202610122020.25

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Solve. List excluded values and check in the original equation. $\frac{4}{2x-2}+\frac{7}{x^2-5x+4}=\frac{3}{x-4}$	1. Excluded values 5 pts Solution/check 15 pts
2. Solve. List excluded values and check candidates in the original equation. $\frac{4}{x^2-4}+\frac{x-3}{x-2}=\frac{5}{x+2}$	2. Excluded values 4 pts Candidates/check 6 pts Solution 10 pts
3. Solve by factoring. $x^2+3x-4=0$	3. Solutions 20 pts
4. Solve by taking square roots; do not expand the square. $2(x+1)^2=8$	4. Solutions 20 pts
5. Solve with the quadratic formula. Simplify radicals. $2x^2+x-2=0$	5. Solutions 20 pts

Instructor answer key: TSM 099 Quadratic and Rational Equations Test

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Excluded values

$\{1, 4\}$

5 pts

Solution/check

$x = 2$

15 pts

2. Excluded values

$\{-2, 2\}$ 4 pts

Candidates/check

$x = 2$ (rejected), $x = 4$ 6 pts

Solution

$x = 4$ 10 pts

3. Solutions

$x = -4,\ 1$

20 pts

4. Solutions

$x = -3,\ 1$

20 pts

5. Solutions

$x = -\frac{1}{4} + \frac{\sqrt{17}}{4},\ \ -\frac{\sqrt{17}}{4} - \frac{1}{4}$ 20 pts